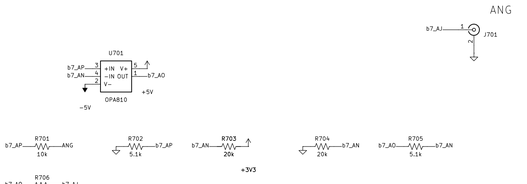
## b7  angle, MT6701

cell (0.0)

tutorial Motors  (+−5 V)

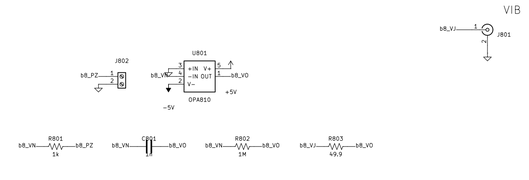



**EXPECT: ANG = a sawtooth, −0.84 V to +0.84 V, once a turn of the shaft (0 at half a turn from the MT6701's zero)**
The MT6701's analog output (0–3.3 V a turn) less 1.65 V, times 0.51, by an OPA810: the shaft's angle without SSI. Its zero is the MT6701's (set by its registers over the SSI, or where the magnet sits).

## b8  vibration, OPA810

cell (1.0)

tutorial Motors  (+−5 V)

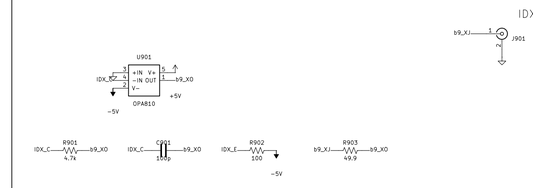



**EXPECT: VIB = −the piezo's charge / 1 nF above 160 Hz: the motor's cogging, its commutation ripple, the bearing**
A piezo disc on the terminal into an OPA810 charge amplifier (1 nF, 1 M: from 160 Hz). Stick the disc to the motor's mount: the cogging at 12 a turn, the commutation's ripple, the bearing, the frame's resonances.

## b9  index, QRE1113

cell (2.0)

tutorial Motors  (+−5 V)

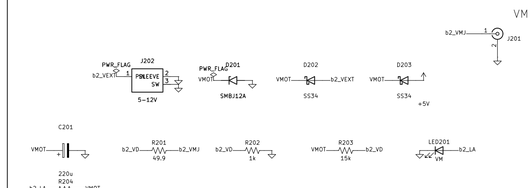



**EXPECT: IDX = 4.7 k × the sensor's current: a pulse once a turn as a white mark on the bell's rim passes**
The QRE1113 under the gimbal's rim (on b5). Its collector held at 0 V by an OPA810: IDX = 4.7 k × Ic.  Put a white mark (tape, paint) on the bell's lower rim: an index pulse a turn, a tachometer with no encoder.

## b2  motor supply, 5–12 V in

cell (3.0)

tutorial Motors

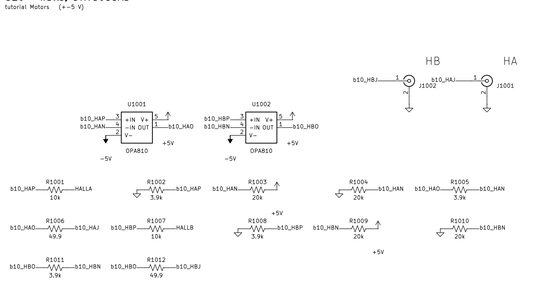



**EXPECT: VMOT = the larger of the barrel jack (5–12 V) and USB 5 V, each less a Schottky's 0.3–0.5 V; VM jack VMOT / 16**
Barrel jack (centre +). 5–12 V, through an SS34 into VMOT, as is USB's 5 V, so VMOT is about 4.6 V with no supply. SMBJ12A and 220 uF on VMOT take the braking energy a motor returns;  VM: VMOT / 16 at high impedance.

## b10  halls, DRV5055A3

cell (0.1)

tutorial Motors  (+−5 V)

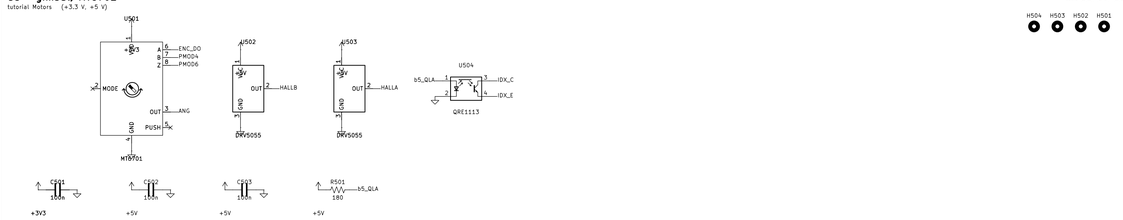



**EXPECT: HA, HB = 9.75 mV/mT from the two halls under the rotor: two sinusoids 120 electrical degrees apart as it turns**
Two DRV5055A3 under the gimbal's magnets (b5); 17.1 degrees apart (120 electrical for 7 pole pairs): each less half its 5 V, times 0.39 (as AMOD−M's m9).  The rotor's electrical angle without the encoder: hall-sensored commutation.

## b5  gimbal, MT6701

cells (1.1), (2.1), (1.2), (2.2)

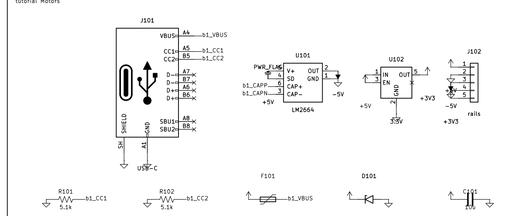
tutorial Motors  (+3.3 V, +5 V)




## b1  power, USB−C 5 V in: +5, −5, +3.3 out

cell (3.1)

tutorial Motors

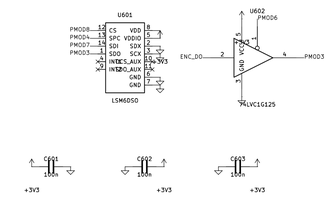



**EXPECT: +5 V from the USB−C (up to 3.5 A), −5 V at 40 mA, +3.3 V at 300 mA; the green LED says 5 V is in, the red one that −5 V is up**
USB−C power only: 5.1 k on each CC asks any source for 5 V.  A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF.
The LM2664 charge pump makes −5 V for the op-amp rails (40 mA; four tiles at 10 mA): it is not switched per tile, an op amp with its +5 V removed draws nothing.  The TPS7A2033 makes 3.3 V for the comparator and the lines that meet the FPGA.
The socket along the top carries +5, ground, −5, ground, +3.3: the bench's rails for a breadboard experiment.

## b6  IMU, LSM6DSO

cell (0.2)

tutorial Motors  (+3.3 V)

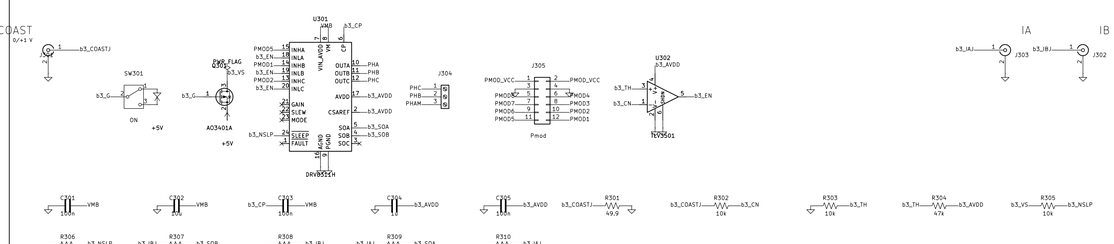



**EXPECT: six axes on SPI (the tutorial's gateware reads them): with the board on edge, the roll about the motor's axis**
LSM6DSO on SPI (mode 3): CS IMU on IO8, SCK IO4, MOSI IO7, MISO IO3, shared with the encoder through a 74LVC1G125 (CS ENC on IO6).  Gyro to 2000 dps, accelerometer to 16 g, to 6.66 kHz.  On edge: a one-axis gimbal's level reference.

## b3  BLDC driver, DRV8311H

cells (1.3), (2.3)

tutorial Motors  (+5 V, VMOT, +−5 V)

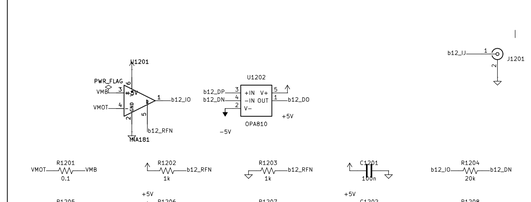



**EXPECT: the motor's phases at the duty cycles of IO5, IO1, IO2 (A, B, C): IA, IB = phase A's and B's currents (low side), 0.75 V/A: COAST high floats them**
DRV8311H in 3× PWM from VMB: INHA IO5, INHB IO1, INHC IO2; EN from COAST through a TLV2501 (high: every phase floats).
IA, IB = SOA, SOB shifted to 0 by 3.3 k / 10 k to −5 V: 0.75 V/A from 2.5 k while each low side conducts.  The motor on the terminals: C, B, A from the left.

## b12  bus current, INA181

cell (3.2)

tutorial Motors  (+−5 V)

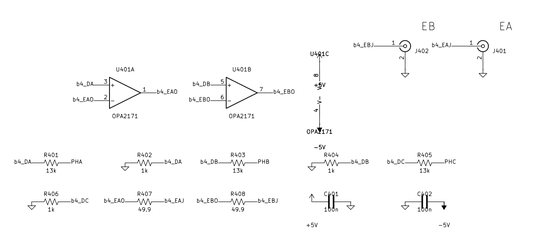



**EXPECT: I = the driver's supply current, 1 V/A (+−1.2 A), negative while the motor brakes**
VMOT through 0.1 ohm to the driver; an INA181A1 (2.5 V on REF) and an OPA810 give I = 1 V/A both ways.  With VM: the electrical power in; against the phase currents: the driver's losses; negative while the motor's energy comes back.

## b4  back−EMF, OPA2171

cell (0.3)

tutorial Motors  (+−5 V)

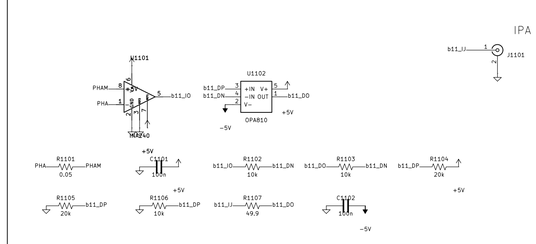



**EXPECT: EA, EB = phase A's and B's voltage / 14 against the neutral: with COAST high and the motor spun, its back−EMF**
Each phase through 13 k / 1 k: with every phase floating the three dividers are the motor's ground, so EA and EB read the back−EMF against the neutral (14 × EA in volts).  Followed by an OPA2171.  A spun gimbal motor makes a few volts.

## b11  phase current, INA240

cell (3.3)

tutorial Motors  (+−5 V)




**EXPECT: IPA = phase A's whole current, 1 V/A both ways, through every PWM state (compare b3's IA: the low side only)**
Phase A through 0.05 ohm between the driver and the motor; an INA240A1 (PWM rejection, common mode to 80 V) about 2.5 V and an OPA810 that takes the 2.5 V away: IPA = 1 V/A, the whole current.  b3's IA sees only the low side's share.

GND  GND
no mounting holes: four stick−on rubber feet, marked on the back

PWR_FLAG
ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/amod_b.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a three−phase driver, a gimbal motor over a magnetic encoder and halls, back−EMF, an IMU
**Board: Jason Gallicchio and Claude  |  open hardware, CERN−OHL−P−2.0  |  github.com/jgallicchio/**

| | | |
|---|---|---|
| Sheet: / | | |
| File: amod_b.kicad_sch | | |
| ***Title: AMOD−B: brushless motors*** | | |
| Size: A1 | Date: 2026−10−25 | **Rev: A** |
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |